

Initial Cell Acquisition

Cell Search & Synchronization (Detect PSS/SSS for timing & cell ID)

nr_initial_sync()

- File: openair1/PHY/NR_UE_TRANSPORT/nr_initial_sync.c
- Link: https://gitlab.eurecom.fr/oai/openairinterface5g/-/blob/develop/openair1/PHY/NR_UE_TRANSPORT/nr_initial_sync.c

Frequency and Time Synchronization (Achieve carrier sync using SSB)

nr_initial_sync() (includes frequency correction)

- File: openair1/PHY/NR_UE_TRANSPORT/nr_initial_sync.c
- Link: https://gitlab.eurecom.fr/oai/openairinterface5g/-/blob/develop/openair1/PHY/NR_UE_TRANSPORT/nr_initial_sync.c

Master Information Block (MIB) Demodulation

PBCH Demodulation (OFDM demodulate PBCH resource elements)

nr_rx_pbch() - PBCH demodulation and soft bit extraction

- File: openair1/PHY/NR_UE_TRANSPORT/nr_pbch.c
- Link: https://gitlab.eurecom.fr/oai/openairinterface5g/-/blob/develop/openair1/PHY/NR_UE_TRANSPORT/nr_pbch.c

nr_get_pbch_re() - PBCH resource element extraction

- File: openair1/PHY/NR_UE_TRANSPORT/nr_pbch.c
- Link: https://gitlab.eurecom.fr/oai/openairinterface5g/-/blob/develop/openair1/PHY/NR_UE_TRANSPORT/nr_pbch.c

Channel Estimation (Use cell-specific reference signals)

nr_dmrs_estimation() - PBCH DMRS processing

- File: openair1/PHY/NR_ESTIMATION/nr_dmrs_rx.c
- Link: https://gitlab.eurecom.fr/oai/openairinterface5g/-/blob/develop/openair1/PHY/NR_UE_ESTIMATION/nr_dl_channel_estimation.c

PBCH Decoding (QPSK demapping, polar decoding, CRC)

nr_pbch_decode() - PBCH decoding wrapper

- File: openair1/PHY/NR_UE_TRANSPORT/nr_pbch.c
- Link: https://gitlab.eurecom.fr/oai/openairinterface5g/-/blob/develop/openair1/PHY/NR_UE_TRANSPORT/nr_pbch.c

nr_pbch_scrambling() - PBCH scrambling/descrambling

- File: openair1/PHY/NR_UE_TRANSPORT/nr_pbch.c
- Link: https://gitlab.eurecom.fr/oai/openairinterface5g/-/blob/develop/openair1/PHY/NR_UE_TRANSPORT/nr_pbch.c

polar_rate_matching() - PBCH rate matching

- File: openair1/PHY/CODING/nrPolar_tools/nr_polar_rate_matching.c
- Link: https://gitlab.eurecom.fr/oai/openairinterface5g/-/blob/develop/openair1/PHY/CODING/nrPolar_tools/nr_polar_procedures.c

polar_decoder() - Polar decoding algorithm

- File: openair1/PHY/CODING/nrPolar_tools/nr_polar_decoder.c
- Link: https://gitlab.eurecom.fr/oai/openairinterface5g/-/blob/develop/openair1/PHY/CODING/nrPolar_tools/nr_polar_decoder.c

MIB ASN.1 Decoding (Extract parameters like SIB1 scheduling, bandwidth parts)

uper_decode_complete() - ASN.1 parsing of BCH payload (MIB)

- File: ASN.1 generated code files
- Directory: openair2/RRC/NR/MESSAGES/
- Link: <https://gitlab.eurecom.fr/oai/openairinterface5g/-/tree/develop/openair2/RRC/NR/MESSAGES/>

System Information Block Type 1 (SIB1) Demodulation

CORESET and PDCCH Detection (Use MIB info to find CORESET#0 & monitor for PDCCH candidates)

nr_coreset_config() - Configure search spaces for DCI

- File: openair1/SCHED_NR_UE/dci_configuration.c
- Link: https://gitlab.eurecom.fr/oai/openairinterface5g/-/blob/develop/openair2/LAYER2/NR_MAC_UE/nr_ue_dci_configuration.c DCI (Downlink Control Information) Decoding (from PDCCH)

nr_dci_decoding_procedure() - DCI decoding for SI-RNTI to find SIB1 PDSCH

- File: openair1/PHY/NR_UE_TRANSPORT/dci_nr.c
- Link: https://gitlab.eurecom.fr/oai/openairinterface5g/-/blob/develop/openair1/PHY/NR_UE_TRANSPORT/dci_nr.c

PDSCH Demodulation (Process PDSCH resources based on DCI)

nr_rx_pdsch() - PDSCH demodulation and decoding

- File: openair1/PHY/NR_UE_TRANSPORT/nr_dlsch_demodulation.c
- Link: https://gitlab.eurecom.fr/oai/openairinterface5g/-/blob/develop/openair1/PHY/NR_UE_TRANSPORT/nr_dlsch_demodulation.c

nr_layer_mapping() - Layer mapping for PDSCH

- File: openair1/PHY/NR_UE_TRANSPORT/nr_dlsch_demodulation.c
- Link: https://gitlab.eurecom.fr/oai/openairinterface5g/-/blob/develop/openair1/PHY/NR_UE_TRANSPORT/nr_dlsch_demodulation.c

Channel Estimation & Equalization (Use DMRS in PDSCH)

nr_pdsch_dmrs_estimation() - PDSCH DMRS processing

- File: openair1/PHY/NR_ESTIMATION/nr_dmrs_rx.c
- Link: https://gitlab.eurecom.fr/oai/openairinterface5g/-/blob/develop/openair1/PHY/NR_UE_ESTIMATION/nr_dl_channel_estimation.c

SIB1 Decoding (LDPC decoding, rate recovery, CRC)

nr_pdsch_scrambling() - PDSCH scrambling/descrambling

- File: openair1/PHY/NR_UE_TRANSPORT/nr_dlsch_demodulation.c
- Link: https://gitlab.eurecom.fr/oai/openairinterface5g/-/blob/develop/openair1/PHY/NR_UE_TRANSPORT/nr_dlsch_demodulation.c

nr_rate_matching_ldpc() - PDSCH rate matching

- File: openair1/PHY/CODING/nrLDPC_decoder/nrLDPC_rate_matching.c

- Link: https://gitlab.eurecom.fr/oai/openairinterface5g/-/blob/develop/openair1/PHY/CODING/nrLDPC_coding/nrLDPC_coding_segment/nr_rate_matching.c

ldpc_decoder() - SIB1 LDPC decode

- File: openair1/PHY/CODING/nrLDPC_decoder/nrLDPC_decoder.c
- Link: https://gitlab.eurecom.fr/oai/openairinterface5g/-/blob/develop/openair1/PHY/CODING/nrLDPC_decoder/nrLDPC_decoder.c

SIB1 ASN.1 Parsing (Extract system configuration like cell access, scheduling, freq info)

nr_rrc_decode_BCCH_DL_SCH_Message() - ASN.1 parsing of SIB1

- File: openair2/RRC/NR/rrc_gNB_NGAP.c or related RRC files
- Link: <https://gitlab.eurecom.fr/oai/openairinterface5g/-/tree/develop/openair2/RRC/NR/>

Essential Initialization Functions

nrLDPC_init() - LDPC decoder initialization

- File: openair1/PHY/CODING/nrLDPC_decoder/nrLDPC_init.c
- Link: https://gitlab.eurecom.fr/oai/openairinterface5g/-/blob/develop/openair1/PHY/CODING/nrLDPC_decoder/nrLDPC_init.c

nr_polar_init() - Polar decoder initialization

- File: openair1/PHY/CODING/nrPolar_tools/nr_polar_init.c
- Link: https://gitlab.eurecom.fr/oai/openairinterface5g/-/blob/develop/openair1/PHY/CODING/nrPolar_tools/nr_polar_init.c